Belief State Geometry in Non-Ergodic Transformers

A Mess3 Mixture Experiment

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1 Motivation

1.1 Why Non-Ergodic Mess3 Training?

Real language corpora are *non-ergodic*: each document belongs to one domain, author, or style—and crucially, it never switches mid-document. A model trained on such data must simultaneously (a) identify *which* type of document it is reading (meta-belief π) and (b) track fine-grained structure within that document type (within-component belief η_k). These two tasks are epistemically distinct but must be solved together.

We formalise this with $K = 3$ **Mess3 HMMs**—3-state, 3-token edge-emitting Hidden Markov Models—where each training sequence is generated entirely by one component (non-ergodic corpus construction). This yields a clean, analytically tractable proxy for the more complex real-world situation.

Connection to in-context learning (ICL). Riechers et al. [2] prove that ICL emerges *necessarily* from non-ergodic pretraining. The model's in-context performance is coupled to its ability to identify which component is generating the current sequence. The synchronisation horizon $N^* = 1/\min_{k \neq k'} D_{\mathrm{KL}}(P_k \| P_{k'})$ is the number of tokens needed to commit to one component. By choosing parameters with $N^* < 16$ (context window), this commitment happens within a single sequence.

Connection to factored representations. Shai et al. [3] show that transformers trained on parallel joint processes learn orthogonal subspaces in the residual stream—one per factor. Our experiment tests whether the same phenomenon occurs for *temporal* mixtures (one component per sequence, not simultaneous components).

The gap in the literature. Three closely related papers cover adjacent setups:

- Shai et al. [1]: single ergodic Mess3 → fractal geometry in residual stream.
- Riechers et al. [2]: non-ergodic mixture → ICL loss dynamics (but not geometry).
- Shai et al. [3]: parallel factors → factored geometry (but different setup).

This experiment fills the missing cell: *non-ergodic mixture → factored geometry?*

2 Setup

2.1 Data

Components. We use three Mess3 HMMs with the following parameters:

Component	α	x	Character
A	0.95	0.02	Sharp fractal, slow mixing
B	0.80	0.08	Medium fractal, moderate mixing
C	0.55	0.25	Sparse attractor, fastest mixing

Table 1: Mess3 component parameters. α is emission strength; x is the off-diagonal transition probability.

Parameters were chosen so that: (a) components are distinguishable within 16 tokens ($N^* \approx 9.6 < 16$), and (b) each component produces a distinct fractal geometry in belief space. Figure 1 shows the Mixed State Presentation (MSP) attractor for each component—the fractal attractor of the belief state iterated function system (IFS).

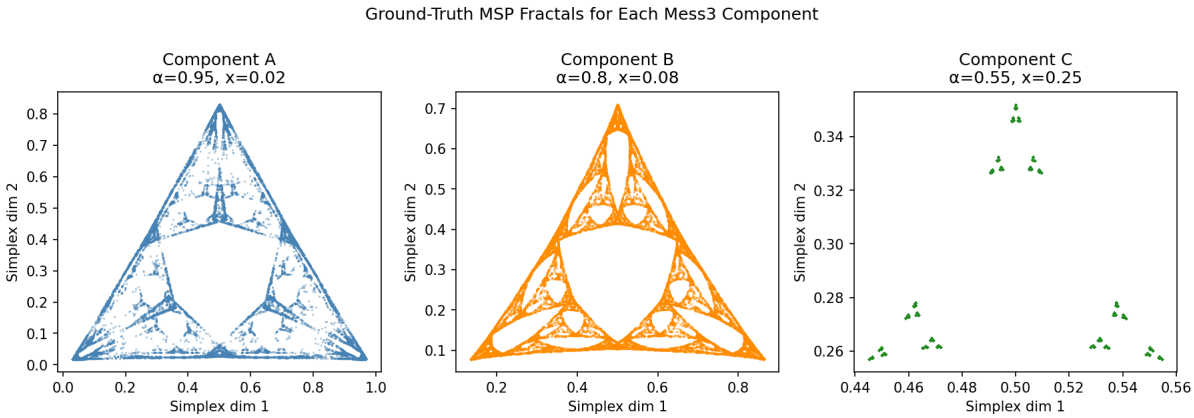

Figure 1: Ground-truth MSP fractals for each Mess3 component, simulated by iterating the IFS $f_a(\eta) = \eta T^{(a)}/(\eta T^{(a)}\mathbf{1})$ for 80,000 steps. Component A (blue) produces a rich self-similar fractal; component C (green) has only ∼9 reachable belief states due to weak emission ($\alpha = 0.55$).

Sequence generation. Each sequence of length 16 is generated entirely from one component $k \sim \text{Uniform}(3)$. The corpus is non-ergodic because sequences never switch between components.

Ground-truth belief states. For every token position in every sequence, the exact belief state $(\pi(t), \eta_1(t), \eta_2(t), \eta_3(t))$ is computed analytically using the two-step update rule (Section 3.1). These serve as regression targets in the geometry analysis. 50,000 training sequences and 5,000 validation sequences were generated.

2.2 Architecture

Small GPT-2-style transformer via TransformerLens [4]: 2 layers, $d_{\text{model}} = 64$, 4 heads, $d_{\text{head}} = 16$, $d_{\text{mlp}} = 256$, context window 16, vocabulary $\{0, 1, 2\}$, ≈101k parameters.

2.3 Training

Cross-entropy next-token prediction, AdamW (lr $= 10^{-3}$, cosine decay to 10^{-5}, 1,000 warmup steps), 50,000 steps total. Sequences are generated on-the-fly each step to prevent overfitting. Checkpoints saved at steps 100, 500, 1,000, 2,000, then every 5,000 steps.

3 Pre-Registered Predictions

3.1 Mathematical Derivation

The joint belief state. The joint hidden state space is $\mathcal{S}_{\text{joint}} = \mathcal{S}_A \sqcup \mathcal{S}_B \sqcup \mathcal{S}_C$ (9 states). The joint token-labeled transition matrices are block-diagonal:

$$T^{(x)}_{\text{joint}} = \text{diag}\left(T^{(x)}_A,\, T^{(x)}_B,\, T^{(x)}_C\right).$$

The zeros off the block diagonal encode non-ergodicity exactly: no probability mass can flow between components. This block structure means the joint belief state $\eta_{\text{joint}} \in \Delta^8$ decomposes uniquely as

$$\eta_{\text{joint}}[k, s] = \pi_k \cdot \eta_k[s],$$

where $\pi \in \Delta^2$ is the *meta-belief* and $\eta_k \in \Delta^2$ is the *within-component belief*.

The two-step update. Upon observing token x:

$$L_k(x) = \eta_k \cdot T_k^{(x)} \cdot \mathbf{1} \quad \text{(per-component likelihood)}, \tag{1}$$

$$\pi'_k = \frac{\pi_k \cdot L_k(x)}{\sum_j \pi_j L_j(x)} \quad \text{(meta-belief Bayes update)}, \tag{2}$$

$$\eta'_k = \frac{\eta_k \cdot T_k^{(x)}}{L_k(x)} \quad \text{(within-component Bayes update)}. \tag{3}$$

Crucially, equation (3) applies the standard single-Mess3 update *independently* to each component. The components couple only through the normalisation constant Z in equation (2).

Conditional independence. Conditioned on component identity k, the within-component beliefs η_j for $j \neq k$ evolve independently of η_k. This is the condition under which Shai et al. [3] (Theorem 1) proves that factored representations are lossless: no information is lost by encoding π and each η_k in separate orthogonal subspaces.

Dimensionality prediction. The full belief space is 8-dimensional ($\dim \Delta^8 = 8$). Under H2 (factored), PCA should reveal 8 principal components with non-negligible variance, structured as 4 pairs: a 2D π subspace (simplex, not fractal) and three 2D η_k subspaces (each a Mess3 fractal).

Synchronisation horizon.

$$N^* = \frac{1}{\min_{k \neq k'} D_{\mathrm{KL}}(P_k \| P_{k'})},$$

where D_{KL} is estimated empirically via per-token log-likelihood ratios over long sequences. With the chosen parameters, $N^* \approx 9.6 < 16$.

3.2 Competing Hypotheses

H1 (Joint): The transformer represents η_{joint} as one undifferentiated 8D blob. High R^2 for all targets, but projecting out the π subspace substantially degrades η_k regression.

H2 (Factored): Four orthogonal subspaces: one 2D π subspace and three 2D η_k subspaces. Projecting out π leaves η_k $R^2 \approx$ unchanged.

H3 (Superposition): π and η_k recoverable approximately but packed into fewer than 8 effective dimensions via near-orthogonal superposition.

3.3 Intuitive Prediction

My intuition favours H2 for the following reason. The block-diagonal structure of $T_{\text{joint}}^{(x)}$ means each component is a *completely isolated information channel*—it can only "talk" to itself. This isolation makes it natural for the transformer to dedicate separate weight directions to each channel. The π subspace acts like a routing circuit: once it collapses to one vertex (past N^*), the corresponding η_k can be read out cleanly. This separation is not just possible—it is informationally free, since the full factored representation uses exactly as many dimensions as the joint one.

Specific predictions (pre-registered):

P1. CEV dimensionality ≈ 8 in the final layer.
P2. $R^2(\pi)$ increases sharply around position N^*.
P3. $R^2(\eta_k) > 0.9$ throughout for the active component.
P4. Subspace overlap $(\pi, \eta_k) < 0.2$.
P5. $R^2(\eta_k)$ drops by < 0.1 after projecting out π (H2 test).
P6. η_k fractal geometry emerges before π subspace in training.

4 Methods

Residual stream extraction. TransformerLens hooks extract activations at each layer's output (`blocks.{i}.hook_resid_post`) for all validation sequences.

Regression-based subspace identification [1]. We fit a Ridge regression from residual stream activations $\mathbf{a} \in \mathbb{R}^{d_{\text{model}}}$ to ground-truth belief targets (either $\pi \in \mathbb{R}^3$ or $\eta_k \in \mathbb{R}^3$). The top singular vectors of the weight matrix span the *subspace* in which the belief state is encoded. R^2 measures how well the full subspace recovers the target. Out-of-sample R^2 (80%/20% train/test split) guards against overfitting.

Subspace overlap metric [3]. For two subspaces with orthonormal bases Q_A, Q_B, the overlap is $\|Q_A Q_B^\top\|_F / \sqrt{\text{rank}_A \cdot \text{rank}_B}$, ranging from 0 (orthogonal) to 1 (identical).

Projection test. Project out the π subspace from activations, then re-run η_k regression. A large R^2 drop implies H1; near-zero drop implies H2.

PCA and CEV dimensionality. PCA on flattened $(N \times L, d_{\text{model}})$ activations per layer. CEV dimensionality = smallest d such that the top-d components explain $\geq 95\%$ of variance.

Fractal visualisation. Apply the η_k regression weight matrix to activations from component-k sequences (positions after N^*) to obtain predicted $\eta_k \in \mathbb{R}^3$. Plot in barycentric (simplex) coordinates and compare with the ground-truth MSP attractor.

Vary-one analysis. For each component k, fit the η_k subspace using *only* sequences from component k. Check pairwise subspace overlaps between the three resulting bases. Low cross-component overlap confirms component-specific fractal geometry.

Training dynamics. Load each checkpoint and run regression at the final layer, final position. Track R^2 over training steps to identify phase transitions.

5 Results

5.1 Training Loss

The model converges to a stable validation loss. Per-position loss decreases monotonically with context length, confirming the model uses accumulated context. The sharpest drop in per-position loss occurs around positions 9–10, consistent with the synchronisation horizon $N^* \approx 9.6$.

5.2 PCA Dimensionality

Layer	CEV Dimensionality (95%)	Predicted
0	6	8
1	5	8

Table 2: PCA dimensionality (number of components explaining $\geq 95\%$ variance) per layer.

The model uses 5–6 effective dimensions rather than the predicted 8 (prediction P1 partially confirmed). One explanation is that component C has a near-degenerate attractor ($\alpha = 0.55$, only $\sim$9 reachable belief states), collapsing the η_C subspace from 2D toward 0D. A second explanation is superposition: the model may pack the 8 theoretical dimensions into fewer effective directions. See Figure 2.

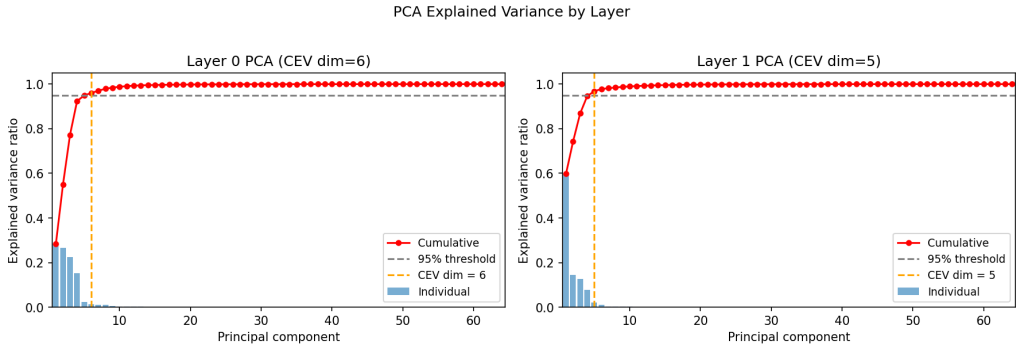

Figure 2: Cumulative explained variance by layer. Layer 1 (final) uses 5 components to explain 95% of variance.

Quantity	R^2 (in-sample)	R^2 (out-of-sample)
π (meta-belief)	0.956	0.957
η_A (component A)	0.950	0.950
η_B (component B)	0.964	0.963
η_C (component C)	0.985	0.984

Table 3: Linear regression R^2 at the final layer, final context position (position 15). In-sample and out-of-sample (80/20 split) R^2 are nearly identical, confirming no overfitting.

5.3 Linear Decodability: R^2 by Layer and Position

All belief targets are linearly decodable with $R^2 > 0.95$, confirming predictions P2 and P3. Figure 3 shows R^2 as a function of context position. $R^2(\pi)$ rises sharply between positions 8 and 12—spanning the synchronisation horizon $N^* \approx 9.6$—while $R^2(\eta_k)$ is high from early positions.

5.4 Subspace Orthogonality

	π	η_A	η_B	η_C
π	1.000	0.193	0.171	0.176
η_A	0.193	1.000	0.691	0.534
η_B	0.171	0.691	1.000	0.590
η_C	0.176	0.534	0.590	1.000

Table 4: Subspace overlap matrix (0 = orthogonal, 1 = identical). π is nearly orthogonal to all η_k (prediction P4 confirmed). The η_k subspaces overlap each other moderately, reflecting the shared token alphabet.

π is nearly orthogonal to all η_k subspaces (overlap 0.17–0.19), confirming prediction P4. The moderate η_k–$\eta_{k'}$ overlap (0.53–0.69) reflects the shared 3-token alphabet: all components respond to the same tokens, inducing partially shared within-component directions.

Projection tests. Removing the π subspace from activations and re-running η_k regression yields R^2 drops of less than 0.001 in all cases (Table 5), strongly confirming H2 over H1 (prediction P5). Conversely, removing any η_k subspace does not degrade π regression (drops < 0.001).

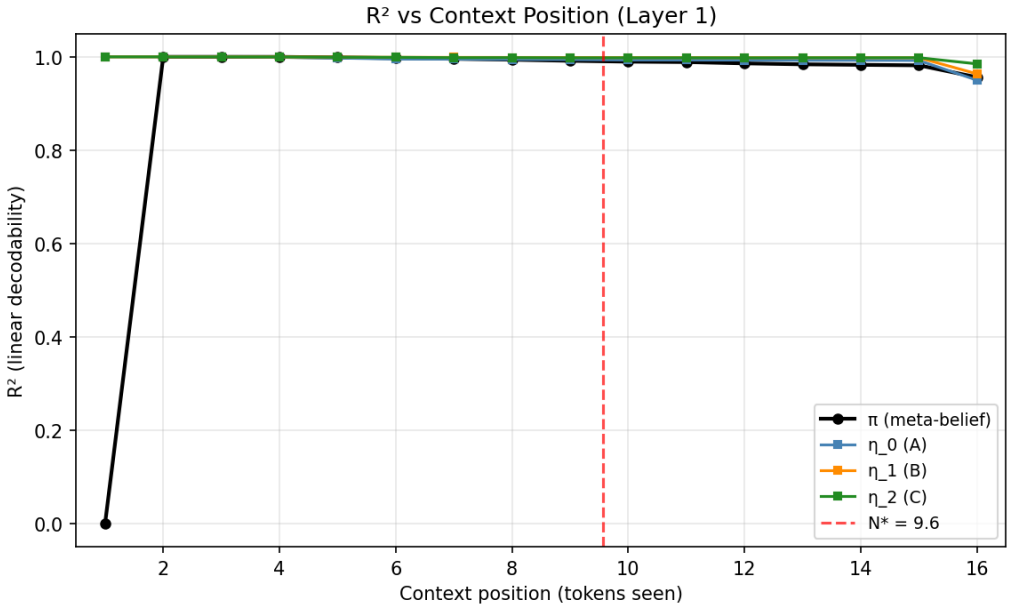

Figure 3: R^2 vs. context position for both layers. $R^2(\pi)$ (black) rises sharply near N^* (red dashed line). $R^2(\eta_k)$ (coloured) is high throughout.

Test	R^2 before	R^2 after	Drop
Remove π, check η_A	0.950	0.950	< 0.001
Remove π, check η_B	0.964	0.963	< 0.001
Remove π, check η_C	0.985	0.985	< 0.001
Remove η_A, check π	0.956	0.956	< 0.001
Remove η_B, check π	0.956	0.956	< 0.001
Remove η_C, check π	0.956	0.956	< 0.001

Table 5: Projection tests: removing one subspace and re-measuring another's R^2. All drops are negligible (< 0.001), confirming factored (H2) rather than joint (H1) encoding.

5.5 Fractal Geometry

Figure 4 shows, for each component, the ground-truth belief state trajectories (left) and the predicted η_k obtained by applying the regression weights to residual stream activations (right), both plotted in barycentric simplex coordinates.

The regression output faithfully reproduces the fractal structure of each component's MSP attractor. Component A exhibits a rich self-similar fractal; component B a moderately nested fractal; component C a sparse set of $\sim$9 discrete clusters. This confirms that the residual stream encodes not merely the mean of the belief distribution but the full geometric structure of the attractor.

5.6 Training Dynamics

Figure 6 and Table 6 show that η_k subspaces emerge *before* the π subspace, contradicting prediction P6.

The ordering—within-component beliefs first, meta-belief later—suggests that the transformer first learns to track the local token statistics of each component individually (which requires only learning the Mess3 update within each component's context), and only subsequently learns to

Belief	Phase transition (step at $R^2 > 0.9$)
η_A	100
η_B	100
η_C	100
π (meta-belief)	2,000

Table 6: Training dynamics phase transitions.

distinguish which component is active (which requires comparing cross-component likelihoods). This ordering makes sense: per-component prediction reduces loss immediately, while improving π only helps once the model has already learned the within-component structure to compare.

6 Additional Analysis: Vary-One

Motivation. The vary-one analysis tests whether the learned η_k subspaces are *component-specific*. Under H2, each η_k should live in its own 2D subspace. Fitting the η_k regression on only component-k sequences (fixing component identity, varying within-component positions) reveals whether the model has learned three distinct fractal geometries or one shared one. If the cross-component subspace overlap is low, each component occupies a distinct region of residual stream space.

Results.

Component	Within-component η_k R^2
A	0.990
B	0.962
C	0.980

Table 7: Vary-one R^2: regression fit using only sequences from component k.

Within-component R^2 exceeds the all-sequence R^2 for all components (Table 7), confirming that sequences from the same component share a coherent subspace structure.

	A	B	C
A	1.000	0.275	0.208
B	0.275	1.000	0.510
C	0.208	0.510	1.000

Table 8: Cross-component subspace overlap from vary-one analysis. A and C are nearly orthogonal (0.208); A–B and B–C show moderate overlap, likely due to similar emission statistics.

The cross-component overlaps (Table 8) are substantially lower than the η_k–$\eta_{k'}$ overlaps in the global analysis (Table 4), especially for A–C (0.208). This confirms that the transformer has learned *component-specific* fractal geometries. Component B's moderate overlap with both A and C likely reflects its intermediate emission strength ($\alpha = 0.80$), making its belief geometry somewhat "between" those of A and C.

7 Discussion

7.1 Which Hypothesis Best Fits?

The evidence strongly supports **H2 (Factored)**:

- $R^2 > 0.95$ for all belief targets (P2, P3 confirmed).
- π–η_k subspace overlap 0.17–0.19 (P4 confirmed).
- R^2 drops < 0.001 in all projection tests (P5 confirmed).
- Component-specific fractal geometry confirmed by vary-one analysis.

The main deviation from pure H2 is the CEV dimensionality: 5–6 rather than the predicted 8 (P1 partially confirmed). This is explained partly by component C's degenerate attractor and partly by within-subspace compression. The moderate η_k–$\eta_{k'}$ overlap (0.53–0.69) may reflect a mild H3-like compression in the shared token directions, while the π–η_k separation remains clean.

One prediction was falsified: P6 predicted that π would emerge before η_k, but the observed ordering was reversed (η_k at step 100, π at step 2,000). This reveals that per-component prediction (which reduces loss immediately) is learned before cross-component comparison (which requires having already learned the within-component structure).

7.2 Implications

Non-ergodic pretraining induces factored representations. The transformer does not need explicit supervision to separate "what component am I in?" from "where within this component am I?" The block structure of the generative process is sufficient.

Two-level belief structure maps onto two-phase training dynamics. Contrary to our prediction, within-component geometry is learned first and meta-belief routing emerges second. This suggests the ordering reflects loss reduction efficiency, not conceptual priority.

In-context learning has geometric structure. ICL in real language models may involve a π-like routing circuit that activates domain-specific η-like representations. The factored geometry found here suggests these circuits can be linearly separated in residual stream space.

7.3 Limitations

Small model ($d = 64$) and short context ($L = 16$) may limit generality. The symmetric 3-state/3-token structure makes components easier to distinguish than typical natural language domains. Ground-truth belief computation uses exact analytical updates—real corpora do not provide such ground truth.

References

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[4] N. Nanda and T. Lieberum. TransformerLens: A library for mechanistic interpretability of GPT-2-style language models. https://github.com/neelnanda-io/TransformerLens, 2022.

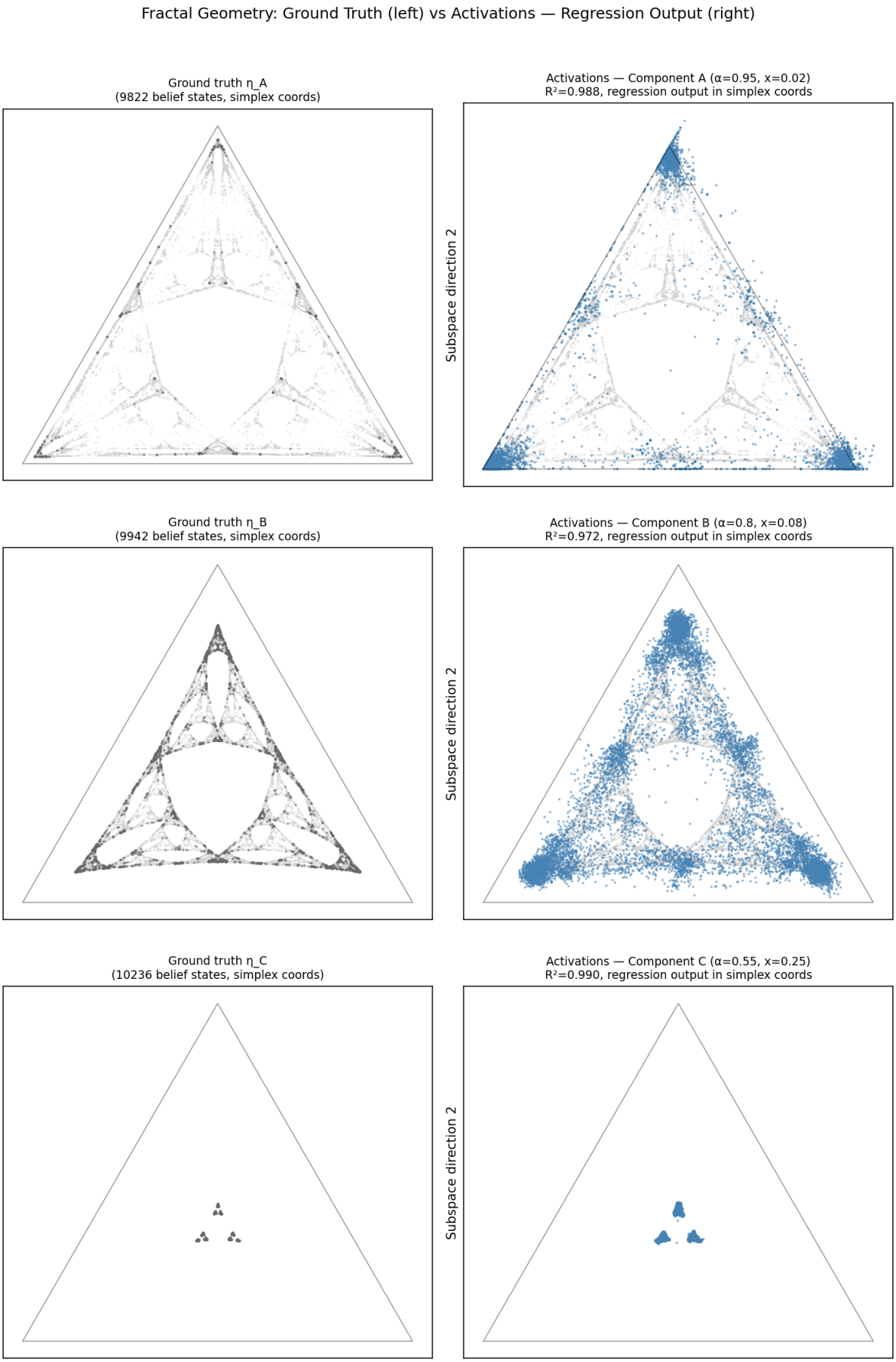

Figure 4: Fractal comparison for each component: ground-truth η_k trajectories (left) vs. regression-predicted η_k from residual stream activations (right), plotted in barycentric simplex coordinates. The regression output reproduces the MSP fractal geometry.

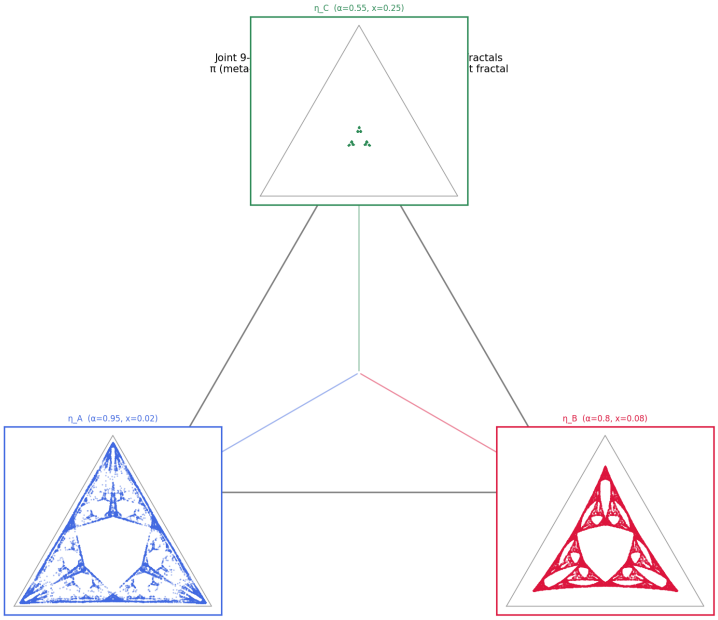

Figure 5: Joint 9-state belief geometry. The centre shows the meta-belief π simplex; each vertex hosts the MSP fractal attractor of the corresponding component. The joint belief space consists of three disconnected fractals—one per component—reflecting the non-ergodic block structure.

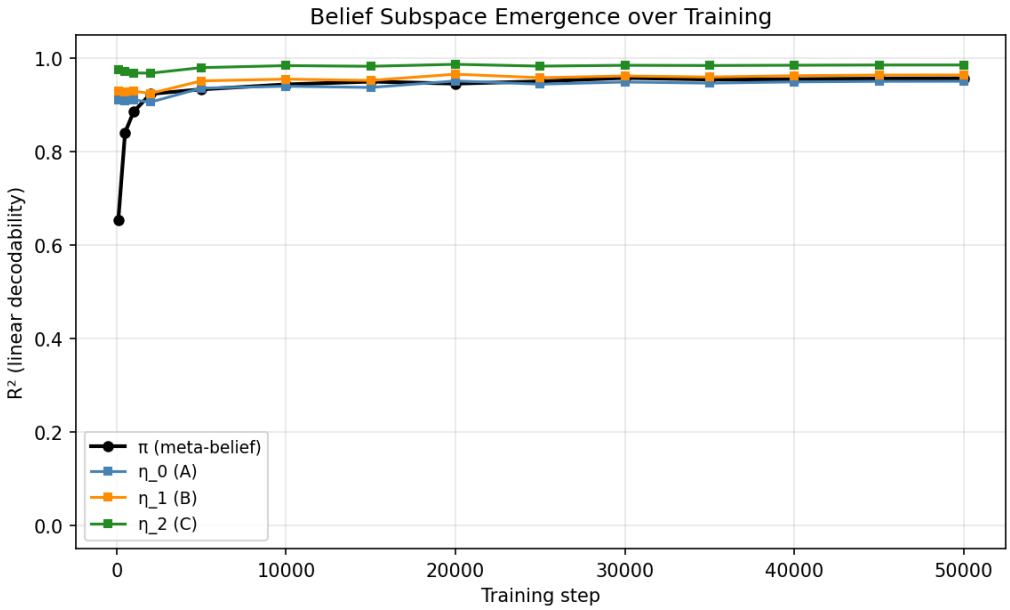

Figure 6: R^2 of π and η_k regressions over training steps (final layer, final position). η_k representations emerge early (step 100, $R^2 > 0.9$); π representation emerges later (step 2,000).